ALL BRANCH (Hinglish)

General Aptitude

Quantitative Aptitude

DPP 03 Discussion Notes

Percentages



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MCQ

The population of a city is increased by 30% and thus became 78000. What is the original population?

$1.3 \times x = 78,000$

$x = \frac{78,000}{1.3}$

$= 60,000$

A 76000

B 64200

C 60000

D None

MCQ

A save 20% of his income. His income is increased by 20% and so he increased his expenditure by 30%. What is the percentage change in his savings?

(A) 20% fall

(B) 4% fall

(C) 20% rise

(D) 4% rise

I	S	Ex
100	20 ↓	80
120	16 ↓	104

$\frac{4}{20} \times \frac{100}{5} = 20\% \downarrow$

MCQ

In an election between two candidates, A and B, A secured 56% of the votes and won by 48000 votes. Find the total number of votes polled if 20% of the votes were declared invalid.

A 500000

B 400000

C 600000

D None

56% — 44%
A B

56 − 44 = 12%

12% of D.V = 48,000

$$D.V = \frac{48,000}{0.12} = 4,00,000$$

$0.8 \times T.Votes = 4,00,000$

$$T.Votes = \frac{4,00,000}{0.8} = 5,00,000$$

MCQ

$\frac{300}{50} = 6/-$

A reduction of 10% in price of sugar enables a housewife to buy 5 kg more for Rs. 300/-. Find the reduced price per kg of sugar.

A 5/-

B 4.5/-

C 6/-

D None

$0.9 \times \frac{300}{x} = \frac{300}{x+5}$

$\frac{270}{x} = \frac{300}{x+5}$

$\Rightarrow 10x = 9x + 45$

$x = 45$

$\frac{30}{5} = 6$

MCQ

In a list of weights of candidates appearing for police selections, the weight of A is marked as 58 kg instead of 46.4 kg. Find the percentage of correction required.

wrong

correct

A 30

B 20

C 24

D None

$$\text{Correction \%} = \frac{\text{diff}}{\text{W. Value}} \times 100$$

$$= \frac{11.6}{58} \times 100 = \frac{1160}{58}$$

$$= 20\%$$

MCQ

In an examination, a student secures 40% and fails by 10 marks. If he scored 50%, he would pass by 15 marks. Find the minimum marks required to pass the exam.

10%

40% ——— 50%

25 marks

10% of Total Mark = 25

Total Mark = 250

125
−15
110

A 250

B 100

C 110

D 125

MCQ

If 240 ltr of oil is poured into a tank, it is still 20% empty. How much more oil is to be poured to fill the tank?

A 300 ltr

B 60 ltr

C 120 ltr

D None

80% of Tank Capacity = 240

÷4 ÷4

20% of Tank Capacity = 60 lit

MCQ

A and B were hired for the same salary. A got two 40% hikes whereas B got a 90% hike. What is the percentage difference in the hikes they got?

(A) 16%

(B) 6%

(C) 10%

(D) 8%

$A \Rightarrow 1.4 \times 1.4 = 1.96$

96% ↑

$B \Rightarrow 90\% \uparrow$

$96 - 90 = 6\%$

MCQ

The population of a town increases at a rate of 10% for every year. If the present population is 12100, find the population two years ago.

A 11000

B 9800

C 10000

D 10120

$1.1 \times 1.1 \times x = 12100$

$1.21 \text{ of } x = 12100$

$x = \frac{12100}{1.21} = 10000$

MCQ

In an election 10 % of the votes were invalid. 40% of the votes were for A and the rest to B. B won with a majority of 243 votes, the total number of votes polled is

A 1250

B 1350

C 1215

D 1336

40% 20% 60%

A B

20% of D.V. = 243

$D.V = \frac{243}{0.2} = \frac{2430}{2}$

= 1215

243

90% of T.P.V = 1215

$T.P.V = \frac{1215}{0.9}$

$= \frac{12150}{9} = 1350$

Thank You!

GW Soldiers